AES Encryption

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Group C

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Goals

- Implement AES 128, 192, and 256 encryption in C
- Measure execution time for repeated encryptions and calculate average time per encryption
- Compare the difference between encryption times based on block sizes
- Evaluate performance of AES software implementation

What is AES

- Advanced Encryption Standard
- AES encrypts data into 16-byte blocks using a key and a series of rounds that transform the data
- This encryption makes the original data unreadable without the correct key
- Common uses include wireless networking, disk encryption, web security, etc.

Benchmarking

Benchmarking is the process of measuring system performance by running a task repeatedly and recording execution time.

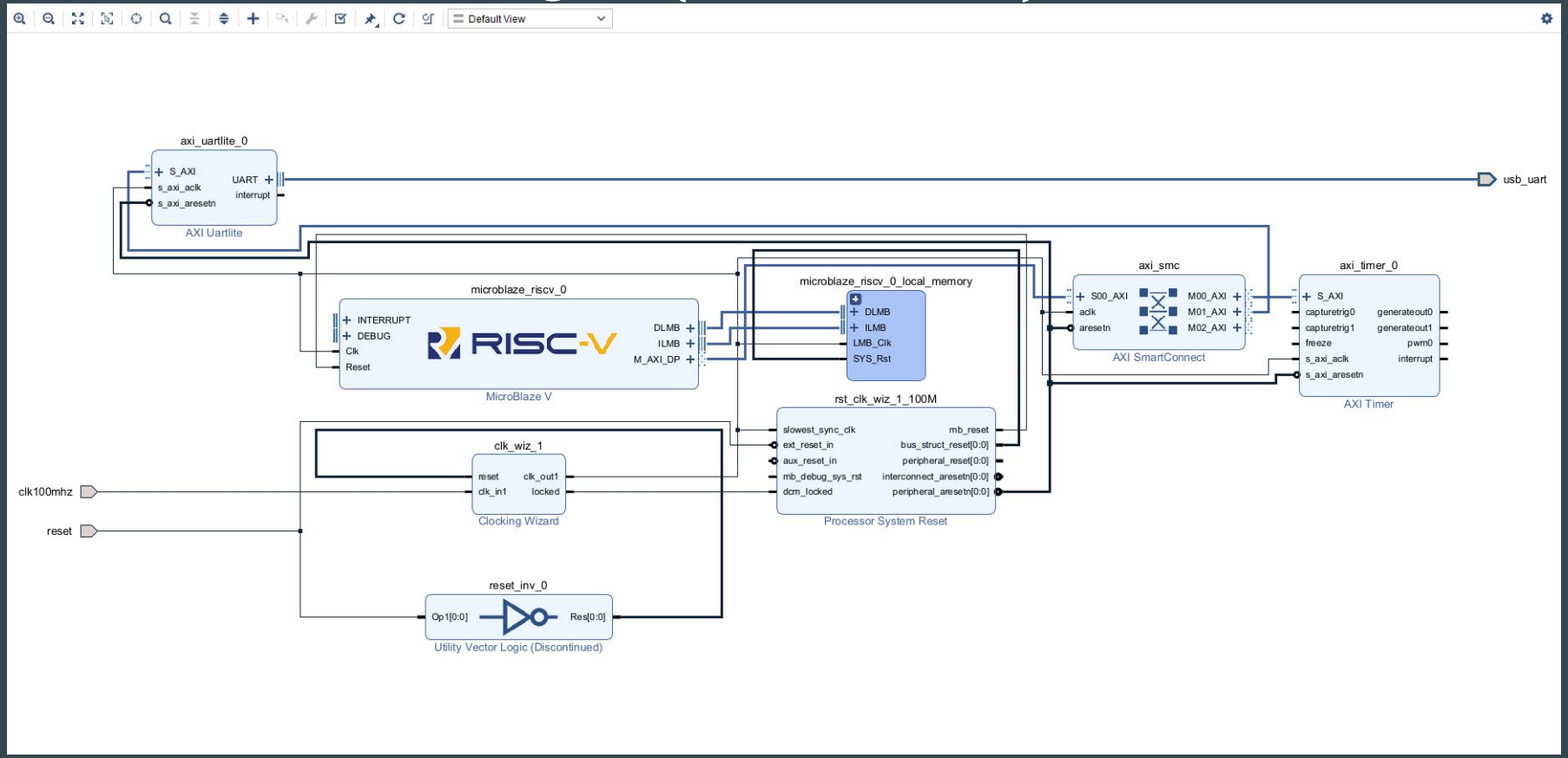
In this project:

- AES encryption is executed many times
- Total runtime is measured
- Average time per encryption is calculated.

System Setup

- Programming Language: C
- AES library implementation
- Standard test vectors for verification
- Repeated encryption loop for benchmarking

Microblaze V Block Diagram (Done in Vivado)



Implementation

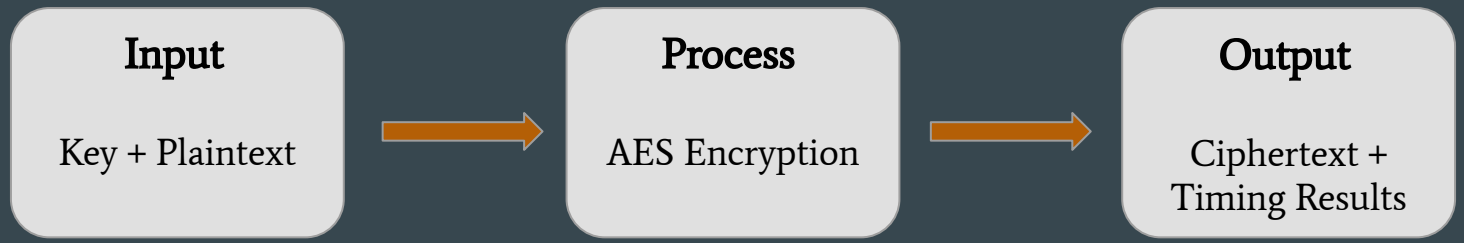
- Define encryption key and plaintext
- Initialize AES context
- Encrypt data block using ECB mode
- Repeat encryption multiple times
- Compare result with expected ciphertext

```c
printf("Testing AES-128\n");
uint8_t key[16] = {
    0x2b,0x7e,0x15,0x16,0x28,0xae,0xd2,0xa6,
    0xab,0xf7,0x15,0x88,0x09,0xcf,0x4f,0x3c
};

// Plaintext
uint8_t plaintext[16] = {
    0x32,0x43,0xf6,0xa8,0x88,0x5a,0x30,0x8d,
    0x31,0x31,0x98,0xa2,0xe0,0x37,0x07,0x34
};
```

```c
uint8_t buffer[16]; //temporary 16 byte
struct AES_ctx ctx; //stores the expand

for (int i = 0; i < numtests; i++) { //
memcpy(buffer, plaintext, 16);
AES_init_ctx(&ctx, key);
AES_ECB_encrypt(&ctx, buffer);
}
```

Program Flow

- Load key and plaintext
- Initialize AES encryption context
- Encrypt 16-byte data block
- Repeat encryption 1 million times
- Measure total execution time
- Compute average time per encryption



Output

- Plaintext = original unscrambled text
- Expected = hard-coded expected output
- Ciphertext = output after many rounds of encryptions
- We did 1 million encryptions each for 128, 192, and 256-bit AES using the same plaintext and measured execution time

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```
AES test starting...
Testing AES-256
PLAINTEXT  : 6bc1bee22e409f96e93d7e117393172a
CIPHERTEXT : f3eed1bdb5d2a03c064b5a7e3db181f8
EXPECTED   : f3eed1bdb5d2a03c064b5a7e3db181f8
TEST PASSED

Ran 1000000 encryptions

AES test starting...
Testing AES-256
PLAINTEXT  : 6bc1bee22e409f96e93d7e117393172a
CIPHERTEXT : f3eed1bdb5d2a03c064b5a7e3db181f8
EXPECTED   : f3eed1bdb5d2a03c064b5a7e3db181f8
TEST PASSED

Ran 1000000 encryptions

AES test starting...
Testing AES-256
PLAINTEXT  : 6bc1bee22e409f96e93d7e117393172a
CIPHERTEXT : f3eed1bdb5d2a03c064b5a7e3db181f8
EXPECTED   : f3eed1bdb5d2a03c064b5a7e3db181f8
TEST PASSED

Ran 1000000 encryptions

AES test starting...
Testing AES-256
PLAINTEXT  : 6bc1bee22e409f96e93d7e117393172a
CIPHERTEXT : f3eed1bdb5d2a03c064b5a7e3db181f8
EXPECTED   : f3eed1bdb5d2a03c064b5a7e3db181f8
TEST PASSED

Ran 1000000 encryptions
```

Tests (Time to complete 1 million encryptions)

	AES-128	AES-192	AES-256
Trial 1	6:32.5	8:19.2	9:08.2
Trial 2	6:32.3	8:19.1	9:08.08
Trial 3	6:32.6	8:19.2	9:08.3
Trial 4	6:32.1	8:19.0	9:07.95
Avg	6:32.375	8:19.125	9:08.1325

Results

- Each run - 1 million encryptions
- AES 128 : 6 min and 32.375 seconds → 392.375 seconds
  - this is about 392.4 μs per encryption, or 2,548.4 encryptions per second
- AES 192: 8 min and 19.125 seconds → 499.125 seconds
  - This is about 499.1 μs per encryption, or 2,003.6 encryptions per second
- AES 256: 9 min and 8.1325 seconds → 548.1325 seconds
  - This is about 548.1 μs per encryption, or 1,824.5 encryptions per second

Extra Tests (Personal computers, AES-256 10,000,000 encryptions)

AMD Ryzen 5 2600 @ 3.4 GHz (32.3 seconds)
AMD Radeon RX 580
16 GB RAM

Apple M1 @ 3.2 GHz (21.4 seconds)
Apple M1 integrated GPU
8 GB RAM

Intel i7-11800H @ 2.30GHz (21.2 seconds)
Intel UHD Graphics
32 GB RAM

AMD Ryzen 9 7940 @ 4.00GHz (20.54 seconds)
Radeon 780M
16 GB RAM

Observations

We were unable to isolate singular variables such as clock speed, amount of RAM, cache, and any hardware AES capability, however:

- Our own computers ran the AES encryptions much faster than our FPGA boards
  - About 2.1 seconds for 1 million AES-256 encryptions vs. 548 seconds for FPGA software-only
- Many CPUs have special hardware dedicated to AES (AES-NI in Intel & AMD CPUs, ARM cryptography extensions, etc. that are much faster than software AES
- Clock speed makes a big difference (100MHz on FPGA vs. 2-4GHz on PCs)
- Running AES in parallel (such as with GPU acceleration) means multiple blocks of AES can be encrypted at the same time, vs. sequential software AES 1 at a time

Conclusion

- AES encryption was successfully implemented in C
- Encryption correctness was verified using known test vectors
- Performance was measured using repeated encryption
- Average time per encryption was calculated
- Results show increased runtime with larger key sizes